

Smart Traffic Signal Control Using Local AI for Emergency Vehicle Route Coordination

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Abstract—Traditional traffic signals commonly operate using predefined timing schedules and may not respond effectively to sudden changes in traffic conditions. Recent research has explored adaptive traffic signal control using artificial intelligence and machine vision. This paper presents a conceptual local-AI traffic signal architecture designed to detect traffic conditions and support emergency vehicle route coordination. The proposed architecture uses street cameras, local computing units, a cloud server for optional storage and analysis, and traffic-light controllers. Local processing is intended to reduce dependence on continuous cloud communication and support rapid decision-making. The proposed system focuses particularly on detecting an emergency vehicle such as an ambulance and coordinating nearby signals to support a clear route. The paper describes the system architecture, methodology, flowchart, algorithm, benefits, limitations, and future scope.

Index Terms—smart traffic signals, local AI, machine vision, emergency vehicle detection, traffic management, adaptive traffic control

I. INTRODUCTION

Traffic signal control plays an important role in managing road traffic and reducing unnecessary delays. Conventional traffic lights generally use predefined timing schedules. Such systems are simple to operate, but they may not respond efficiently when traffic conditions change unexpectedly.

Artificial intelligence (AI) and machine vision have created new possibilities for adaptive traffic signal control. Research has examined AI-based traffic signal control, including approaches based on deep reinforcement learning [1].

Machine-vision techniques have also been investigated for adaptive traffic-light control by using information obtained from road scenes [2].

These approaches indicate the potential of using real-time traffic information instead of relying only on fixed schedules.

Cloud-based systems can provide centralized computing and storage. However, continuous communication with a remote server may introduce communication dependency and delay. A local-AI approach moves part of the processing closer to the traffic signal, allowing road information to be processed locally.

This paper proposes a conceptual smart traffic signal architecture that combines local traffic analysis with optional cloud connectivity. The main focus is emergency vehicle route coordination, particularly for ambulances.

II. PROBLEM STATEMENT

Traffic conditions can change rapidly because of congestion, accidents, road closures, or emergency situations. A fixed traffic signal schedule cannot automatically respond to all such changes.

An emergency vehicle such as an ambulance may need to travel through several signalized intersections. If the signals continue operating according to their normal schedules, the vehicle may experience unnecessary delays.

Therefore, there is a need for a traffic management approach that can use real-time road information and support coordination of nearby traffic signals during emergency situations.

III. OBJECTIVES

The objectives of the proposed system are:

- 1) To monitor traffic conditions using road-side cameras.
- 2) To process selected traffic information locally using a computing unit.
- 3) To identify an emergency vehicle from available traffic information.
- 4) To coordinate nearby traffic signals when an emergency vehicle is detected.
- 5) To reduce dependence on continuous cloud communication for immediate decisions.
- 6) To provide a scalable architecture for future intelligent traffic-management applications.

IV. RELATED WORK

AI-based traffic signal control has been studied using different machine-learning approaches. Rasheed et al. reviewed deep reinforcement learning methods for traffic signal control and discussed their application to changing traffic conditions [1].

Machine vision provides another approach for obtaining information about vehicles and road conditions. Gorodokin et al. investigated optimization of adaptive traffic-light control modes using machine vision [2].

These studies provide a foundation for exploring traffic signal systems that use real-time information. However, the specific emergency-vehicle coordination architecture presented in this paper is a proposed system design rather than a reported experimental result.

V. PROPOSED SYSTEM

The proposed system combines local traffic analysis with traffic-signal control.

The architecture consists of four major components:

- 1) Street Cameras
- 2) Smart Pole Computer
- 3) Optional City Cloud Server
- 4) Traffic-Light Controller

The street cameras capture traffic information. The smart pole computer performs local processing of the relevant information. Selected information may be forwarded to a cloud server for storage or wider analysis. The traffic-light controller receives the required control decision.

The conceptual information flow is:

Street Cameras → Smart Pole Computer → Cloud Server → Traffic Lights

For emergency response, local processing can be used to identify the emergency-vehicle condition and initiate the appropriate signal-coordination procedure.

VI. SYSTEM ARCHITECTURE

The proposed architecture is shown in Fig. 1.

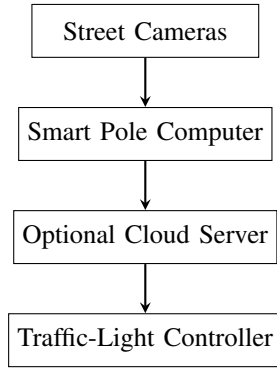


Fig. 1. Conceptual Architecture of the Proposed System

The local computer acts as the primary processing component for immediate traffic-related decisions. The cloud component is optional and can be used for storage, monitoring, and large-scale analysis.

VII. SYSTEM FLOWCHART

The proposed workflow is shown in Fig. 2.

VIII. METHODOLOGY

The proposed methodology consists of the following stages.

A. Traffic Data Collection

Street cameras collect visual information from the road. The collected information may include the presence and movement of vehicles.

B. Local Processing

The smart pole computer processes the relevant camera information locally. The purpose of local processing is to support immediate decision-making without requiring every operation to be performed by a remote server.

C. Emergency Vehicle Detection

The system analyzes the available traffic information to determine whether an emergency vehicle is present. In a practical implementation, this module would require a validated computer-vision or machine-learning model.

D. Signal Coordination

When an emergency vehicle is detected, the system determines which nearby signals should be coordinated to support the vehicle's route.

E. Traffic Signal Control

The selected traffic signals receive the control instruction. After the emergency vehicle has passed, the signals can return to their normal operating mode.

IX. PROPOSED ALGORITHM

The following algorithm describes the conceptual operation of the proposed system.

Algorithm 1 Emergency Vehicle Signal Coordination

- 1: Start
 - 2: Capture traffic information
 - 3: Process information using local computing
 - 4: Detect emergency vehicle
 - 5: **if** emergency vehicle is detected **then**
 - 6: Determine required route
 - 7: Identify nearby traffic signals
 - 8: Generate signal-coordination instructions
 - 9: Send instructions to selected signals
 - 10: Monitor the emergency route
 - 11: **else**
 - 12: Continue normal traffic signal operation
 - 13: **end if**
 - 14: Repeat monitoring
-

X. DATA REPRESENTATION

The system can represent traffic information using a simple data structure.

A traffic observation can be represented as:

$$T = (V, L, E, S) \quad (1)$$

where:

- V represents the detected vehicles,
- L represents the traffic location,
- E represents the emergency-vehicle status,
- S represents the current signal state.

A simplified traffic-state vector can be represented as:

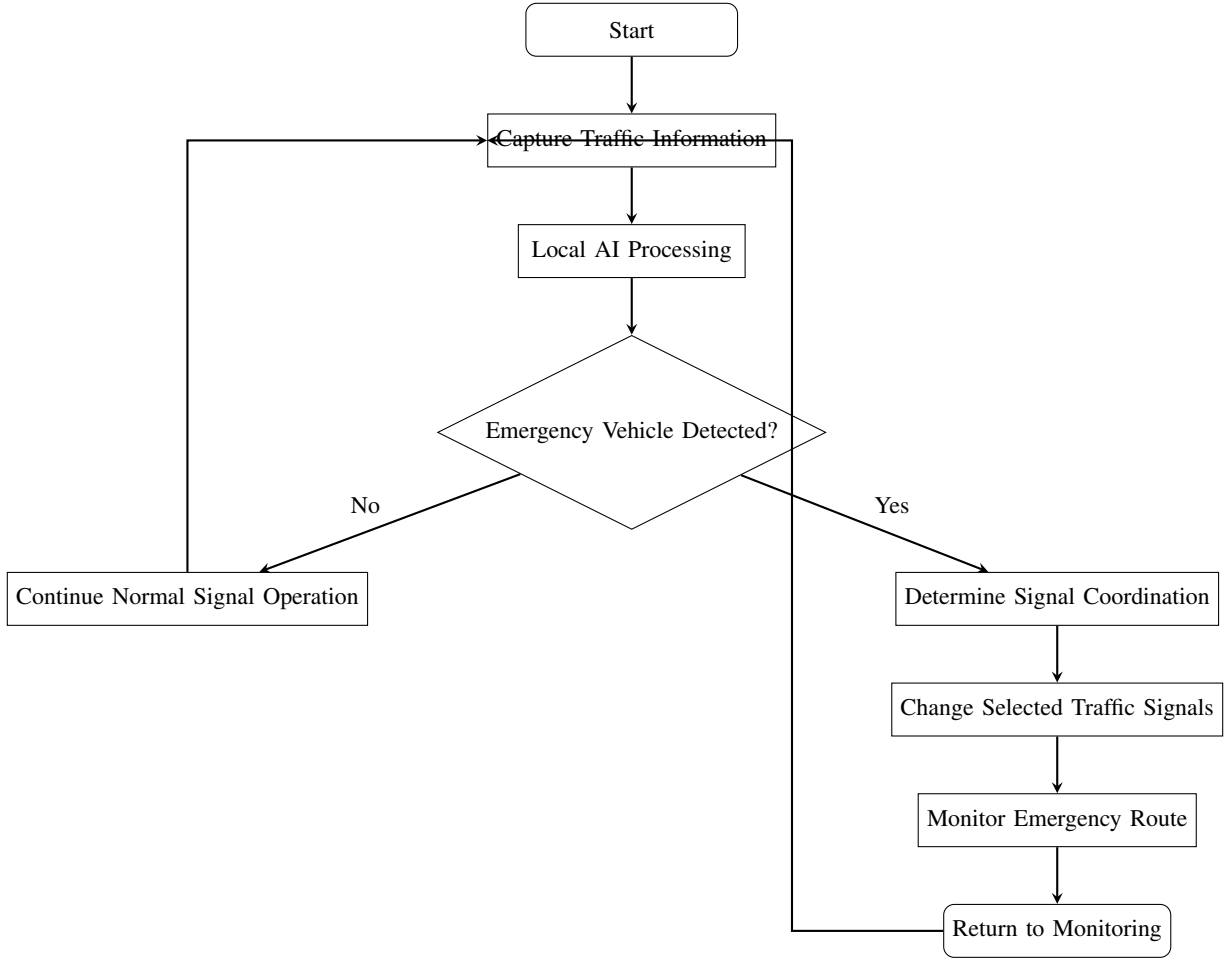


Fig. 2. Flowchart of the Proposed Emergency Vehicle Traffic-Signal Coordination System

$$X = [V_1 \ V_2 \ V_3 \ E]$$

where V_1 , V_2 , and V_3 represent traffic information from selected road segments and E represents the emergency condition.

XI. PERFORMANCE MEASURES

A practical implementation can be evaluated using measurable performance indicators.

One possible measure is emergency-route response time:

$$T_r = T_c + T_p + T_d + T_s \quad (2)$$

where:

- T_c is the camera capture time,
- T_p is the local processing time,
- T_d is the decision time,
- T_s is the signal-control communication time.

The actual values of these parameters must be obtained through experiments using a physical or simulated system. No experimental values are assumed in this paper.

XII. SAMPLE TEST PLAN

The following table provides a test-plan structure for a future prototype evaluation.

TABLE I
PROPOSED SYSTEM TEST CASES

Test	Condition	Expected Behavior
T1	Normal traffic	Normal signal operation
T2	Emergency vehicle detected	Signal coordination initiated
T3	No emergency vehicle	No emergency override
T4	Camera input unavailable	System reports input failure
T5	Emergency condition ends	Signals return to normal operation

These are proposed test cases, not experimentally measured

results.

XIII. ADVANTAGES

The proposed architecture has several potential advantages:

- It can use real-time traffic information.
- Local processing can reduce dependence on continuous cloud communication.
- The architecture can support emergency-vehicle coordination.
- The system can be expanded to multiple intersections.
- Cloud connectivity can be used for storage and large-scale analysis.

These are architectural benefits of the proposed design and would require experimental validation before quantitative performance claims are made.

XIV. LIMITATIONS

The proposed system has several limitations.

- Reliable emergency-vehicle detection requires a properly trained and validated vision model.
- Camera performance may be affected by weather, lighting, occlusion, and camera position.
- Incorrect detection could result in inappropriate signal control.
- Real-world deployment requires integration with existing traffic-control infrastructure.
- Communication and cybersecurity requirements must be addressed before practical deployment.
- The present paper describes a proposed architecture and does not claim measured field performance.

XV. APPLICATIONS

The proposed architecture can potentially be applied to:

- Ambulance route coordination
- Fire-service vehicle coordination
- Police emergency response
- Smart-city traffic management
- Adaptive traffic signal control
- Traffic monitoring and analysis

XVI. FUTURE SCOPE

Future work can focus on developing and experimentally evaluating the proposed architecture.

A computer-vision model can be trained and tested for emergency-vehicle detection. The system can then be evaluated using a traffic simulator or controlled prototype.

Future research can also investigate:

- Multi-intersection coordination
- Vehicle-to-infrastructure communication
- Edge-AI hardware implementation
- Real-time traffic prediction
- Secure communication between traffic controllers
- Performance comparison with fixed-time signals
- Field testing under controlled conditions

XVII. CONCLUSION

This paper presented a conceptual smart traffic signal architecture using local AI for traffic monitoring and emergency vehicle route coordination. The architecture combines street cameras, local processing, optional cloud connectivity, and traffic-light controllers.

Existing research demonstrates interest in AI-based traffic signal control and machine-vision-based adaptive traffic management [1], [2].

Building on these research directions, the proposed system focuses on the specific problem of coordinating traffic signals when an emergency vehicle requires a clear route.

The proposed architecture provides a basis for future prototype development. However, its detection accuracy, response time, reliability, and effectiveness must be established through controlled experiments before any quantitative performance claims can be made.

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